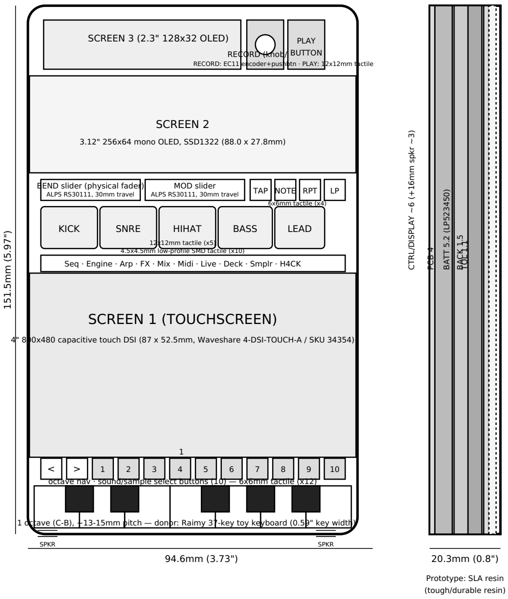
cy83rd3ck — MULTI-SCREEN ENCLOSURE CONCEPT (FROM SKETCH)



Three-screen + physical controls layout (refined per latest sketch)

- **Screen 3** (top-left): small status/transport/waveform display. Top-right: a rotary encoder (with knob) and a second small display/button (OLED B).
- **Screen 2**: secondary mid display (sliders/status overview).
- **Physical row 1**: BEND + MOD sliders (faders), plus TAP / NOTE / RPT / LOOP buttons.
- **Physical row 2**: 5 track-select buttons — KICK / SNRE / HIHAT / BASS / LEAD.
- **Tab row**: Seq/Engine/Arp/FX/Mix/Midi/Live/Deck/Smplr/H4CK — physical buttons or thin touch strip.
- **Screen 1** (large): main 4" 480x800 capacitive touch DSI — sequencer grid, sampler, deck, mixer etc., depending on active tab.
- **Bottom**: physical FSR keyboard, 14 white / 10 black, C4-C6 (unchanged from earlier blueprints).

Big implications:

1. **Displays/drivers**: Screen1 = DSI (CM4). Screen2/3/OLED B = small SPI/I2C displays — CM4 has spare SPI + I2C, workable. Encoder = rotary + push, needs 3 GPIO/Teensy pins.
2. **Physical controls**: BEND/MOD as real sliders + TAP/NOTE/RPT/LOOP/track buttons + tab row is a LOT of physical I/O — likely needs the Teensy 4.1 (or an I2C GPIO expander) to handle button matrix + 2 analog faders + encoder, in addition to the 24-key FSR matrix.
3. **Software**: cyberdeck_v5.tsx currently renders BEND/MOD/transport/track-select/tabs as on-screen UI in one continuous page. This concept moves most of that to physical controls + small secondary displays — meaning Screen1's web UI loses its top control zone entirely, and a separate lightweight renderer (Python/SPI) drives Screen2/3/OLED B from app state.
4. **Thickness**: each extra display + button/fader layer adds depth. Sliders alone are often 6-8mm deep. This pushes well past the 20mm target — likely 24-28mm again, possibly more with fader hardware.

Suggest next step: this is now closer to a full hardware-control-surface redesign than an enclosure tweak. Worth listing exact part numbers for: 2x linear sliders, rotary encoder, ~8 tactile buttons, Screen2/3 panels — then a stepped cross-section.

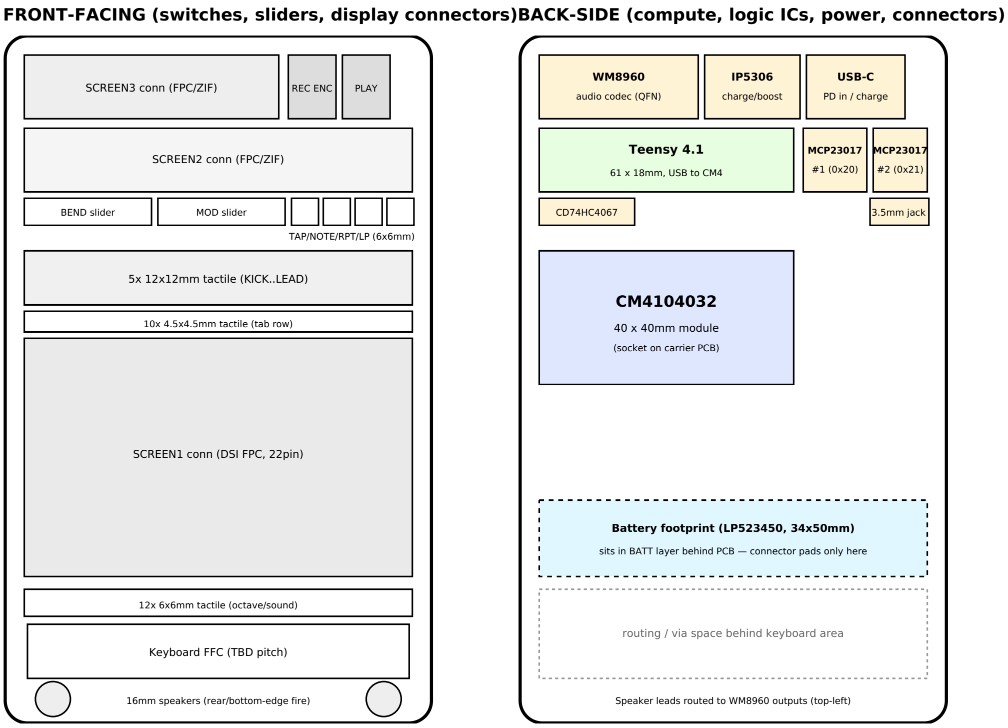
cy83rd3ck	**TITLE**	Multi-Screen Enclosure Concept (3 displays + 2 OLEDs)	**DWG NO.**	CY83-ENC-010 (concept)
	BASIS	88.9 x 127.0mm (3.5 x 5.0in) front face, layout per hand sketch	**SCALE**	AS NOTED
	STATUS	CONCEPT — part numbers / thicknesses / driver routing TBD	**REV**	A — June 2026

cy83rd3ck — BILL OF MATERIALS (PROTOTYPE)

Category	Component	Part / Spec	Notes
Compute	SBC	Raspberry Pi CM4104032 — 4GB RAM / 32GB eMMC / Wireless ($120)	Runs Chromium kiosk + cy83rd3ck app
Compute	Input MCU	Teensy 4.1	USB MIDI controller for all physical inputs
Display	Screen 1	4" 800x480 capacitive touch DSI panel (87 x 52.5mm active area, Waveshare 4-DSI-TOUCH-A / SKU 34354, $28.99)	Main touchscreen UI
Display	Screen 2	3.12" SSD1322 mono OLED, 256x64 (88.0 x 27.8mm)	Waveform display, SPI
Display	Screen 3	2.3" SSD1306 mono OLED, 128x32 (~56.6 x 14.1mm)	Status/transport, SPI/I2C
Controls	BEND / MOD sliders (x2)	ALPS RS30111, 10K linear, 30mm travel	Analog → Teensy ADC
Controls	TAP/NOTE/RPT/LP (x4)	6x6mm tactile switch	Direct GPIO
Controls	Track buttons (x5)	12x12mm tactile switch	Direct GPIO
Controls	Tab row (x10)	4.5x4.5mm low-profile SMD tactile	MCP23017 #1 (0x20)
Controls	Octave nav + sound select (x12)	6x6mm tactile switch	MCP23017 #2 (0x21)
Controls	RECORD encoder	EC11 rotary encoder w/ pushbutton	Direct GPIO (A/B/push)
Controls	PLAY button	12x12mm tactile switch	MCP23017 #1
Controls	Keyboard (1 oct, 7W+5B)	Donor: Raimy 37-key toy keyboard (~13-15mm pitch), FSR-based	FSR matrix → CD74HC4067 mux → Teensy A0
Audio	Codec	WM8960 (I2S + I2C)	Stereo speaker amp + 3.5mm jack + mic in
Audio	Speakers (x2)	BeStar 16mm round, 0.7-1.5W, 4-8Ω (~2.5-3mm thick)	Rear/bottom-edge grilles
Power	Battery	LP523450 LiPo, 1000mAh, 3.7V (5.2 x 34 x 50mm)	
Power	Charge/Boost	IP5306-based charge/boost, USB-C, 5V/2.1A	Powers CM4 + Teensy
Logic	GPIO expanders (x2)	MCP23017, addr 0x20/0x21, SSOP-28	I2C, shared bus w/ Teensy
Logic	Analog mux	CD74HC4067, SOIC-24	Keyboard FSR matrix
Connectors	USB-C	16-pin SMD mid-mount receptacle	PD charging
Connectors	3.5mm jack	PJ-320A-class slim SMD (~6mm depth)	WM8960 headphone out
Connectors	Screen 1 FPC	22-pin, 0.5mm pitch (CM4 DSI)	
Connectors	Screen 2/3 FPC + ZIF	0.5mm pitch FPC, matching ZIF sockets	
Connectors	Keyboard ribbon	TBD — pending donor keybed inspection	
Enclosure	Case (front/back)	SLA resin, tough/durable resin, 1.5mm walls	20.3mm (0.8") total depth

Open items: keyboard ribbon connector (depends on donor unit), final PCB layout integrating all of the above.

cy83rd3ck — PCB FLOORPLAN (CONCEPT, NOT ROUTED)



Notes: This is a single PCB with components on both sides — front side carries switch/slider footprints and display FPC/ZIF connectors aligned to the front-panel layout; back side carries the CM4 module socket, Teensy 4.1, audio codec, power management, GPIO expanders/mux, and the USB-C/3.5mm connectors (positioned at the top edge for enclosure cutout access). The CM4 (40x40mm) is placed behind the track-button row where there's a large open area; Teensy 4.1 (61x18mm) sits behind Screen 2. The battery sits in its own depth layer behind the PCB — only its connector pads are on the PCB, in the area behind Screen 1's lower portion. Keyboard ribbon connector placement is TBD pending donor inspection.

Next step for an actual routable board: import this floorplan into KiCad, place real footprints (CM4 castellated-edge socket, Teensy headers, SSOP/QFN/SOIC packages for the ICs, FPC/ZIF connectors, switch/slider THT or SMD footprints), then route power (5V/3.3V rails) and signal nets (SPI, I2C, GPIO matrix, analog) per the pin-wiring table.

cy83rd3ck — FRONT PANEL SHELL (OpenSCAD source)

First print target: a flat 1.5mm front panel (94.6 x 151.51mm) with all cutouts derived from the layout above. Open in OpenSCAD (free, openscad.org) to render and export STL for SLA printing. Use this to validate button/screen/keyboard alignment against sourced parts before modeling the full enclosure (walls, back, standoffs).

```
// ============================================================
// cy83rd3ck — FRONT PANEL SHELL (concept, prototype)
// Overall: 94.6 x 151.51 x 1.5mm, corner radius 7mm
// Open this file in OpenSCAD (free, openscad.org) to view/export STL.
// All cutout positions derived from the front-panel diagram
// (cy83rd3ck_hardware_concept.pdf), 1 unit = 0.3534mm.
// ============================================================

W = 94.6;
H = 151.51;
T = 1.5;      // panel thickness
R = 7;        // corner radius
EXTRA = 4;    // extra height for cutout tools (through-cuts)

module rrect(w, h, r) {
    hull() {
        translate([r, r])           circle(r=r, $fn=48);
        translate([w-r, r])         circle(r=r, $fn=48);
        translate([r, h-r])         circle(r=r, $fn=48);
        translate([w-r, h-r])       circle(r=r, $fn=48);
    }
}

module panel() {
    linear_extrude(height=T) rrect(W, H, R);
}

module cut_rect(x, y, w, h) {
    translate([x, y, -EXTRA/2]) cube([w, h, T+EXTRA]);
}

module cut_circle(cx, cy, d) {
    translate([cx, cy, -EXTRA/2]) cylinder(d=d, h=T+EXTRA, $fn=48);
}

module front_panel() {
    difference() {
        panel();

        // ---- Display windows ----
        cut_rect(3.29, 78.61, 87, 52.5);     // SCREEN1 (4" 800x480 touch DSI, Waveshare 4-DSI-TOUCH-A / SKU 34354 active area)
        cut_rect(8.91, 24.43, 76.778, 19.178); // SCREEN2 (3.12" SSD1322 OLED active area)
        cut_rect(4.51, 4.11, 56.6, 14.1);    // SCREEN3 (2.3" 128x32 OLED)

        // ---- Record encoder + Play button (top row) ----
        cut_circle(68.11+3.5, 11.16+3.5, 7);  // RECORD encoder shaft, d=7mm
        cut_rect(73.89, 5.16, 12, 12);        // PLAY button

        // ---- TAP / NOTE / RPT / LP (6x6mm) ----
        cut_rect(63.80, 49.84, 6, 6);
        cut_rect(70.88, 49.84, 6, 6);
        cut_rect(77.97, 49.84, 6, 6);
        cut_rect(85.06, 49.84, 6, 6);

        // ---- Track buttons KICK/SNRE/HIHAT/BASS/LEAD (12x12mm) ----
        cut_rect(5.84,  57.63, 12, 12);
        cut_rect(22.80, 57.63, 12, 12);
        cut_rect(39.77, 57.63, 12, 12);
        cut_rect(56.38, 57.63, 12, 12);
        cut_rect(72.02, 57.63, 12, 12);

        // ---- Tab row x10 (4.5x4.5mm) ----
        for (i = [0:9]) {
            cut_rect(5.83 + i*8.726, 71.64, 4.5, 4.5);
        }

        // ---- Octave nav (<,>) + sound select x10 (6x6mm) ----
        oct_x = [3.71, 11.11, 18.50, 25.90, 33.30, 40.70, 48.10, 55.50, 62.89, 70.28, 77.69, 85.07];
        for (x = oct_x) {
            cut_rect(x, 129.77, 6, 6);
        }

        // ---- BEND / MOD slider slots (30mm travel x 4mm) ----
        cut_rect(2.98,  50.84, 30, 4);
        cut_rect(32.93, 50.84, 30, 4);

        // ---- Keyboard cutout (1 octave, FSR keybed pokes through) ----
        cut_rect(1.80, 137.69, 91.0, 12.05);

        // ---- Speaker grilles (bottom edge, simple slot vents) ----
        for (gx = [0:2]) {
            cut_rect(2.83 + gx*5,  H-4, 3, 3); // left grille, 3 slots
            cut_rect(75.77 + gx*5, H-4, 3, 3); // right grille, 3 slots
        }
    }
}

front_panel();

// ============================================================
// NOTES:
// - This is the FRONT PANEL only (1.5mm) — a good first print to
//   validate button/screen/keyboard alignment against real parts.
// - Side walls + back panel + internal standoffs are NOT modeled
//   yet; this is a flat test plate.
// - BEND/MOD slots assume the slider lever protrudes through a
//   30mm slot; verify against the actual ALPS RS30111 lever travel.
// - Track button & octave/sound cutouts are sized to the SWITCH
//   (12x12 / 6x6mm), not a larger keycap — add keycap clearance
//   if using oversized caps.
// - Keyboard cutout is a single rectangle; the FSR keybed assembly
//   sits behind it. Adjust once the donor keyboard is measured.
// ============================================================
```